

Targeted SAT Math — Set 06

Student: Karina

Homework after Bluebook Practice Test 8 and Suite Hard Module 02. Work on paper.

Calculator / Desmos is fine.

These are **new** questions on the items that missed — not those official questions, and not copies from Sets 01–05.

SPR = student-produced response. Write a number (integer, decimal, or fraction).

On #11, write the sign of a , of b , and of c , and a one-line why for each, **before** you pick a choice.

Write the job first.

1. Notes for this set only

Percent less / percent greater

“ $p\%$ less than R ” means the new amount is $R\left(1 - \frac{p}{100}\right)$. It is not $R \cdot \frac{p}{100}$.

“ $p\%$ greater than C ” means the new amount is $C\left(1 + \frac{p}{100}\right)$.

A sale price can be chained: first take the percent off the regular price, then set that sale equal to “percent greater than cost.” Do the two jobs in order.

Histogram bins — smallest and largest mean

The bars only tell you **how many** values sit in each interval, not **which** integers. Inside a bin you may place every integer at the left end or the right end.

- **Smallest** |mean A – mean B |: if the two histograms have the **same** frequencies, you can use the **same** 23 (or 25) integers in both sets. The difference of the means can be **0**.
- **Largest** difference: push set A as far right as the bins allow, and set B as far left (or the other way). Integers from 20 up to but not including 30 go from 20 to 29. The gap in every bin is 9, so the means differ by 9.

Equilateral height, and isosceles right

An equilateral triangle with side s has height $\frac{s\sqrt{3}}{2}$. If they write that height as $k\sqrt{3}$, then $k = \frac{s}{2}$.
Perimeter $3s$ gives s first.

An **isosceles right** triangle has two equal legs and a right angle between them. Legs s, s , hypotenuse $s\sqrt{2}$. The altitude to the hypotenuse is $\frac{s}{\sqrt{2}} = \frac{s\sqrt{2}}{2}$.

Factor $x + 2b$

$(x + 2b)$ is a factor of a quadratic only if $x = -2b$ makes the quadratic 0. Plug $x = -2b$ into each choice. You need a **positive integer** b . A leftover that forces $b = 0$ or a non-integer is dead.

Sphere volume (reference)

$$V = \frac{4}{3}\pi r^3$$

The SAT prints this on the reference page. Cube the radius, then multiply. If they ask for V/π , the π drops.

Percent of last year vs percent of the start

- Each year the value increases by $p\%$ **of its value the previous year** $\Rightarrow$ multiply by $1 + \frac{p}{100}$ every year $\Rightarrow$ **increasing exponential**.
- Each year it increases by $p\%$ **of the initial amount** $\Rightarrow$ add the **same** dollar amount every year $\Rightarrow$ **increasing linear**.

Read which amount they percented. Decreasing versions flip the sign. Linear vs exponential is “add a constant” vs “multiply by a constant.”

Scatterplot $y = ax^2 + bx + c$

- Sign of a :** opens down $\Rightarrow a < 0$. Opens up $\Rightarrow a > 0$.
- Sign of c :** the y -intercept. Above the origin $\Rightarrow c > 0$. Below $\Rightarrow c < 0$.
- Sign of b :** the vertex is at $x = -\frac{b}{2a}$. If that x is **positive** and $a < 0$, then b must be **positive**.
If that x is **negative** and $a < 0$, then b is **negative**.

Write the three signs before you hunt the matching equation.

Quadratic sold-vs-price — minimum is still a vertex

“Minimum of k sold at price h ” is the vertex (h, k) , and $a > 0$. One more point finds a .

$$N(p) = a(p - h)^2 + k$$

The graph is symmetric about $p = h$. Prices the same distance from h sell the **same** number. A line through two prices is the wrong model.

Volunteer poll vs random sample

A website / call-in / “viewers go to our site” poll is **not** a random sample. Percents that add to 98 instead of 100 do **not** kill the poll.

If they selected people **at random from a whole population**, you may generalize to that **whole population** (the sampling frame). A table that later shows only a subset does not shrink the frame.

Random assignment of a treatment is an **experiment** and can support a cause. A survey that just watches who already does what is **observational** and cannot.

Point-slope constant is $F(h)$, not $F(0)$

$$F(x) = A + m(x - h)$$

The printed A is the value when $x = h$. Convert h back to a date or a clock time. $F(0)$ is a different plug-in: $A + m(0 - h) = A - mh$.

Exponential: $p\%$ less than the last value

$$f(x) = A(r)^x, \quad 0 < r < 1$$

Then $f(n) = r \cdot f(n - 1)$, so $f(n)$ is $(1 - r) \cdot 100\%$ **less** than $f(n - 1)$. The leftover r (for example 0.28) is **not** p . $p = 72$, not 28.

2. Questions

1. (SPR) The regular price of a jacket at a store is \$18.00. The sale price of the jacket is 75% less than the regular price, and the sale price is 25% greater than the store's cost for the jacket. What was the store's cost, in dollars, for the jacket?

2. The scatterplot of temperature y , in degrees Fahrenheit, x hours after sunrise is well modeled by a parabola. A few of the points are $(0, 52)$, $(2, 68)$, $(5, 80)$, $(8, 68)$, and $(10, 52)$. Of the following equations, which best models the data?

- A) $y = 0.8x^2 + 8x - 52$
- B) $y = -0.8x^2 - 8x + 52$
- C) $y = 0.8x^2 + 8x + 52$
- D) $y = -0.8x^2 + 8x + 52$

3. Near the end of a city-council livestream, the host invited viewers to answer a poll on the show's website: "Do you support the new parking rule discussed tonight?" The host later reported that 31% answered "Yes" and 66% answered "No." Which of the following best explains why the results are unlikely to represent the residents of the city?

- A) The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.
- B) Those who responded to the poll were not a random sample of the residents of the city.
- C) There were not 50% "Yes" responses and 50% "No" responses.
- D) The show did not allow viewers enough time to respond to the poll.

4. (SPR) The perimeter of an equilateral triangle is 468 centimeters. The height of this triangle is $k\sqrt{3}$ centimeters, where k is a constant. What is the value of k ?

5. (SPR) A certain mug was sold online for prices varying between \$6 and \$14. The number of mugs sold can be modeled by a quadratic function based on the price, where 7 mugs were sold for \$8 each and the **minimum** of 3 mugs were sold for \$12 each. According to the model, what is the number of mugs that were sold for \$16 each?

6. Each year, the value of an account increases by 0.37% of its value the previous year. Which of the following best models how the value of the account changes over time?

- A) Decreasing exponential
- B) Decreasing linear

- C) Increasing exponential
- D) Increasing linear

7. A museum's ticket price, in dollars, w weeks after May 1 is modeled by

$$C(w) = 36.50 - 0.80(w - 6).$$

The constant 36.50 in this function estimates which of the following?

- A) The average weekly decrease in the ticket price
- B) The difference in the ticket price from May 1 to June 12
- C) The ticket price on May 1
- D) The ticket price on June 12

8. (SPR) Two data sets of 25 integers each are summarized by the same frequencies.

Interval	Frequency
$0 \leq x < 10$	5
$10 \leq x < 20$	6
$20 \leq x < 30$	7
$30 \leq x < 40$	7

The first interval is the integers greater than or equal to 0 and less than 10, and so on. What is the **smallest** possible difference between the mean of data set A and the mean of data set B ?

9. (SPR) A solid sphere has radius 6. Its volume is given by $V = \frac{4}{3}\pi r^3$. What is the value of $\frac{V}{\pi}$?

10. A botanist wants to know whether a new fertilizer changes tomato yield. She has 40 plants of the same variety. She assigns 20 plants at random to the new fertilizer and 20 to the old fertilizer, grows them in the same greenhouse, and compares the mean yield. Which statement is true?

- A) This is an observational study, so a difference in mean yield cannot be used to claim the fertilizer caused the change.
- B) This is an experiment with random assignment, so a difference in mean yield can be used to claim the fertilizer caused the change.

- C) The results can be generalized to all plants of every species, because 40 is a large sample.
D) The study is biased because the two groups do not have the same number of plants.

11. A scatterplot of y versus x is modeled by $y = ax^2 + bx + c$. The parabola opens **downward**, crosses the y -axis **below** the origin, and has its vertex at a **negative** x -value.

Write the sign of each coefficient and a one-line why. Then pick the matching choice.

- Sign of a : _____
Why: _____
- Sign of b : _____
Why (use $x = -\frac{b}{2a}$): _____
- Sign of c : _____
Why: _____

Which of the following could be the signs of a , b , and c ?

- A) $a < 0$, $b > 0$, $c < 0$
B) $a < 0$, $b < 0$, $c < 0$
C) $a > 0$, $b < 0$, $c < 0$
D) $a < 0$, $b < 0$, $c > 0$

12. Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

- A) $3x^2 + 10x + 20b$
B) $3x^2 + 19x + 20b$
C) $3x^2 + 25x + 20b$
D) $3x^2 + 34x + 20b$

13. (SPR) The perimeter of an isosceles right triangle is $40 + 20\sqrt{2}$ centimeters. The altitude to the hypotenuse is $k\sqrt{2}$ centimeters, where k is a constant. What is the value of k ?

14. A clinic director selected 80 patients at random from **all** patients at the clinic. The table shows only the 35 patients in that sample who had at least 4 visits last year.

	4 to 6 visits	7 or more visits	Total
Improved	12	9	21
Did not improve	8	6	14
Total	20	15	35

Which of the following is the largest population to which the results of the study can be generalized?

- A) All clinic patients who improved
- B) The 80 patients in the sample
- C) The 35 patients in the sample who had at least 4 visits
- D) All patients at the clinic

15. The function f is defined by $f(x) = 35(0.28)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. What is the value of p ?

- A) 28
- B) 35
- C) 65
- D) 72

16. Each year, the value of an account increases by 0.37% of the **initial** investment (the amount on day one), not of last year's balance. Which of the following best models how the value of the account changes over time?

- A) Decreasing exponential
- B) Decreasing linear
- C) Increasing exponential
- D) Increasing linear

17. (SPR) A bakery sold a loaf online at prices from \$4 to \$12. The number sold can be modeled by a quadratic function based on the price. The **minimum** of 2 loaves were sold for \$9 each, and 6 loaves were sold for \$5 each. According to the model, what is the number of loaves that were sold for \$13 each?

18. The height, in centimeters, of a sunflower t hours after 6:00 a.m. is modeled by

$$H(t) = 142 + 3.5(t - 8).$$

The constant 142 in this function estimates which of the following?

- A) The average hourly increase in height
- B) The change in height from 6:00 a.m. to 2:00 p.m.
- C) The height at 6:00 a.m.
- D) The height at 2:00 p.m.

19. (SPR) Two data sets of 23 integers each are summarized by the same frequencies.

Interval	Frequency
$20 \leq x < 30$	3
$30 \leq x < 40$	4
$40 \leq x < 50$	7
$50 \leq x < 60$	9

The first interval is the integers greater than or equal to 20 and less than 30, and so on. What is the **greatest** possible difference between the mean of data set *A* and the mean of data set *B*?

20. A school counselor notices that students who already eat breakfast in the cafeteria have higher math-quiz averages than students who do not. She did not assign anyone to eat or skip breakfast. Which statement is true?

- A) This is an experiment, so she can conclude that eating breakfast caused the higher averages.
- B) This is an observational study, so she cannot conclude that eating breakfast caused the higher averages.
- C) The finding can be generalized to all teenagers in the country, because a school is a random sample of the country.
- D) The two groups must have the same number of students, or the comparison is invalid.